

OpenStax Calculus Volume 2

Section 3.7 Exercises: Improper Integrals

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Evaluate the following integrals. If the integral is not convergent, answer “divergent.”

1. $\int_2^4 \frac{dx}{(x-3)^2}$
2. $\int_0^\infty \frac{1}{4+x^2} dx$
3. $\int_0^2 \frac{1}{\sqrt{4-x^2}} dx$
4. $\int_1^\infty \frac{1}{x \ln x} dx$
5. $\int_1^\infty x e^{-x} dx$
6. $\int_{-\infty}^\infty \frac{x}{x^2+1} dx$
7. Without integrating, determine whether the integral $\int_1^\infty \frac{1}{\sqrt{x^3+1}} dx$ converges or diverges by comparing the function $f(x) = \frac{1}{\sqrt{x^3+1}}$ with $g(x) = \frac{1}{\sqrt{x^3}}$.
8. Without integrating, determine whether the integral $\int_1^\infty \frac{1}{\sqrt{x+1}} dx$ converges or diverges.

Determine whether the improper integrals converge or diverge. If possible, determine the value of the integrals that converge.

9. $\int_0^\infty e^{-x} \cos x dx$
10. $\int_1^\infty \frac{\ln x}{x} dx$
11. $\int_0^1 \frac{\ln x}{\sqrt{x}} dx$
12. $\int_0^1 \ln x dx$
13. $\int_{-\infty}^\infty \frac{1}{x^2+1} dx$
14. $\int_1^5 \frac{dx}{\sqrt{x-1}}$
15. $\int_{-2}^2 \frac{dx}{(1+x)^2}$
16. $\int_0^\infty e^{-x} dx$
17. $\int_0^\infty \sin x dx$

18. $\int_{-\infty}^{\infty} \frac{e^x}{1+e^{2x}} dx$
19. $\int_0^1 \frac{dx}{\sqrt[3]{x}}$
20. $\int_0^2 \frac{dx}{x^3}$
21. $\int_{-1}^2 \frac{dx}{x^3}$
22. $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$
23. $\int_0^3 \frac{1}{x-1} dx$
24. $\int_1^{\infty} \frac{5}{x^3} dx$
25. $\int_3^5 \frac{5}{(x-4)^2} dx$

Determine the convergence of each of the following integrals by comparison with the given integral. If the integral converges, find the number to which it converges.

26. $\int_1^{\infty} \frac{dx}{x^2+4x}$; compare with $\int_1^{\infty} \frac{dx}{x^2}$.
27. $\int_1^{\infty} \frac{dx}{\sqrt{x+1}}$; compare with $\int_1^{\infty} \frac{dx}{2\sqrt{x}}$.

Evaluate the integrals. If the integral diverges, answer “diverges.”

28. $\int_1^{\infty} \frac{dx}{x^e}$
29. $\int_0^1 \frac{dx}{x^{\pi}}$
30. $\int_0^1 \frac{dx}{\sqrt{1-x}}$
31. $\int_0^1 \frac{dx}{1-x}$
32. $\int_{-\infty}^0 \frac{dx}{x^2+1}$
33. $\int_{-1}^1 \frac{dx}{\sqrt{1-x^2}}$
34. $\int_0^1 \frac{\ln x}{x} dx$
35. $\int_0^e \ln(x) dx$
36. $\int_0^{\infty} xe^{-x} dx$
37. $\int_{-\infty}^{\infty} \frac{x}{(x^2+1)^2} dx$
38. $\int_0^{\infty} e^x dx$

Evaluate the improper integrals. Each of these integrals has an infinite discontinuity either at an endpoint or at an interior point of the interval.

39. $\int_0^9 \frac{dx}{\sqrt{9-x}}$
40. $\int_{-27}^1 \frac{dx}{x^{2/3}}$
41. $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$

42. $\int_6^{24} \frac{dt}{t\sqrt{t^2-36}}$
43. $\int_0^4 x \ln(4x) dx$
44. $\int_0^3 \frac{x}{\sqrt{9-x^2}} dx$
45. Evaluate $\int_{.5}^1 \frac{dx}{\sqrt{1-x^2}}$. (Be careful!) (Express your answer using three decimal places.)
46. Evaluate $\int_1^4 \frac{dx}{\sqrt{x^2-1}}$. (Express the answer in exact form.)
47. Evaluate $\int_2^\infty \frac{dx}{(x^2-1)^{3/2}}$.
48. Find the area of the region in the first quadrant between the curve $y = e^{-6x}$ and the x -axis.
49. Find the area of the region bounded by the curve $y = \frac{7}{x^2}$, the x -axis, and on the left by $x = 1$.
50. Find the area under the curve $y = \frac{1}{(x+1)^{3/2}}$, bounded on the left by $x = 3$.
51. Find the area under $y = \frac{5}{1+x^2}$ in the first quadrant.
52. Find the volume of the solid generated by revolving about the x -axis the region under the curve $y = \frac{3}{x}$ from $x = 1$ to $x = \infty$.
53. Find the volume of the solid generated by revolving about the y -axis the region under the curve $y = 6e^{-2x}$ in the first quadrant.
54. Find the volume of the solid generated by revolving about the x -axis the area under the curve $y = 3e^{-x}$ in the first quadrant.

The Laplace transform of a continuous function over the interval $[0, \infty)$ is defined by $F(s) = \int_0^\infty e^{-sx} f(x) dx$ (see the Student Project). This definition is used to solve some important initial-value problems in differential equations, as discussed later. The domain of F is the set of all real numbers s such that the improper integral converges. Find the Laplace transform F of each of the following functions and give the domain of F .

55. $f(x) = 1$
56. $f(x) = x$
57. $f(x) = \cos(2x)$
58. $f(x) = e^{ax}$
59. Use the formula for arc length to show that the circumference of the circle $x^2 + y^2 = 1$ is 2π .

A nonnegative function is a probability density function if it satisfies the following definition: $\int_{-\infty}^\infty f(t) dt = 1$. The probability that a random variable x lies between a and b is given by $P(a \leq x \leq b) = \int_a^b f(t) dt$.

60. Show that $f(x) = \begin{cases} 0 & \text{if } x < 0, \\ 7e^{-7x} & \text{if } x \geq 0 \end{cases}$ is a probability density function.
61. Find the probability that x is between 0 and 0.3. (Use the function defined in the preceding problem.) Use four-place decimal accuracy.